

Deep Generative Models

Chapter 3: Generation by Explicit Distribution Learning

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Latent Space: Motivation

As we have seen several times: although data samples are **high-dimensional**, their **valid cases** are significantly **limited within the data space**

samples might be thought as if **processed** from a **much easier origin**

- + What do you mean by an **easy origin**?
- We have seen such things a lot! An example makes life easier

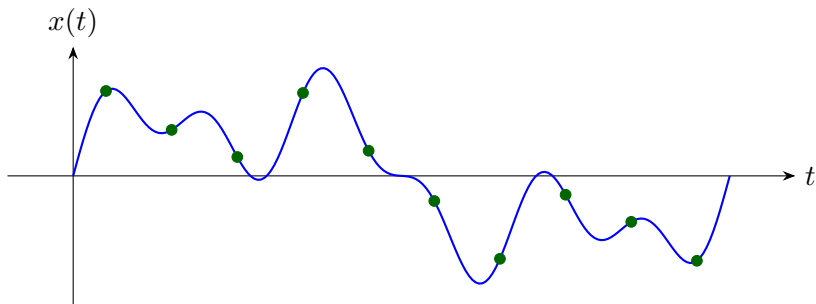
Example: We listen to a set of **audio signals**. We know that these signals have at most **m known harmonics**, i.e.,

$$x(t) = \sum_{j=1}^m z_j \sin(2\pi f_j t)$$

where z_j is the amplitude of harmonic j . We want to learn data distribution.

Latent Representation: *Example*

To make data samples, we could work directly in *time domain*: we *sample* $d \gg 1$ *time samples* according to Nyquist theorem

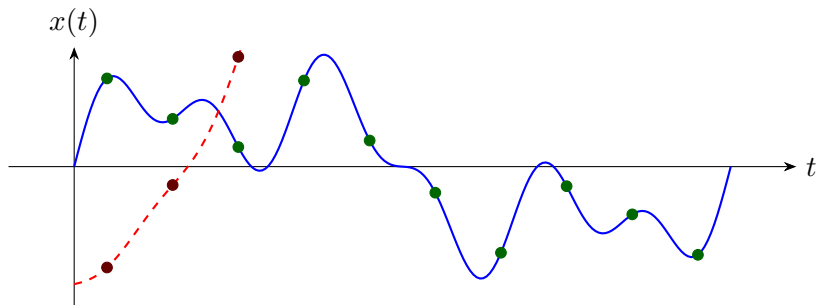


So, *data-space in this case is $\mathbb{R}^d$ with d being potentially very large!*

$$x = [x_1, x_2, \dots, x_d]$$

Latent Representation: *Example*

We however know that many combinations of samples are *impossible!*



For instance, though *data-space* is $\mathbb{R}^d$, samples of *exponential-like signals* are not happening $\rightsquigarrow$ *invalid* data samples!

Latent Representation: *Example*

An alternative way of describing data distribution is to learn the distribution of *harmonic amplitudes*, i.e.,

$$z = [z_1, \dots, z_m]$$

If we know $P(z)$: we can sample $z \sim P(z)$ and generate an audio signal as

$$x(t) = \sum_{j=1}^m z_j \sin(2\pi f_j t)$$

This gives a very simple example of what is known as

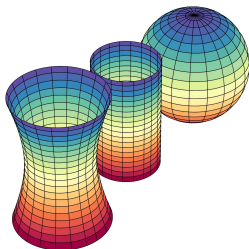
latent representation of data

Data Manifold

Valid data is typically **concentrated** on a

narrow (low-dimensional) manifold hidden in the **data-space**

- + What do you mean by a **manifold**?!
- Think of a spherical surface: *it looks 3D, but we can only move 2D when we are on it* $\rightsquigarrow$ *it's a 2D manifold embedded in 3D space*



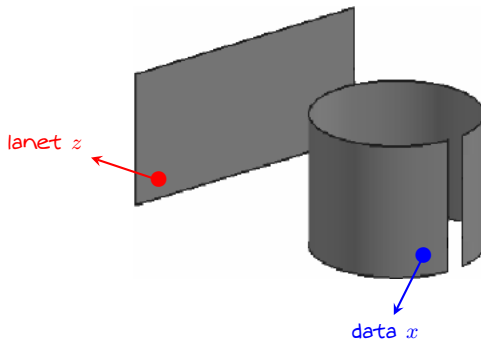
Valid data-points are also the same

- ↳ They are **high-dimensional**
- ↳ They lie on a **thin manifold**

Latent Space

Latent Space

Latent space can be thought of as a coordinate system for (or a transformed version of) *data manifold*, potentially in *lower dimensions*



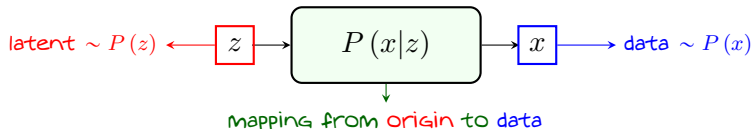
Working with Latent Representation

Considering the notion of **latent space**: in most cases, it's more efficient to work with **latent representations** of data

- Many times, it's easier to model the distribution of **latent representation**
 - ↳ In audio example: assuming **time samples in x** being **i.i.d.** is completely non-sense; however, having **i.i.d. latent z** could make sense!
- Typically, **latent representation** is much lower in dimensions than **data**
 - ↳ MNIST images could have as small as **8D latent space**
- But, we could have **latent representation** of the **same** dimension
 - ↳ In this case, the **latent space** deforms the **complicated data manifold** into a **simpler manifold** in the same space
 - ↳ e.g., a **complicated 2D surface in 3D** is transformed to a **simple 2D plane in 3D**

Latent Representation: *Statistical Interpretation*

From statistical viewpoint: we can look at the *latent representation* as the root which has been *processed* to a visible form of *data*



In this formulation, we can say

$$P(z, x) = P(x|z) P(z) \rightsquigarrow P(x) = \sum_z P(z, x) = \sum_z P(x|z) P(z)$$

Attention

There is *no* unique latent space: we have various (potentially infinite) choices for *latent space* $\rightsquigarrow$ each choice has its own $P(z)$ and *mapping* $P(x|z)$

Normalizing Flow: A Simple Probability Problem

Assume that we have the continuous **random variable** z

$$z \sim Q(z)$$

We pass it through **function** $f : \mathbb{R} \mapsto \mathbb{R}$ and compute x

$$x = f(z)$$

with f having the following properties

- f is **invertible**, i.e., we can find $z = f^{-1}(x)$
- f is **strictly increasing**, i.e., $z_1 < z_2 \rightsquigarrow f(z_1) < f(z_2)$
 - ↳ This concludes that f^{-1} is also **strictly increasing**

We want to find the **density of** x

Change of Variable

Let's start with computing the cumulative distribution of x

$$\begin{aligned}\text{CDF}_x(a) &= \Pr\{x \leq a\} = \Pr\{f(z) \leq a\} \\ &= \Pr\{z \leq f^{-1}(a)\} = \text{CDF}_z(f^{-1}(a))\end{aligned}$$

By definition, the *distribution of x* is the derivative of CDF, i.e.,

$$\begin{aligned}\text{density of } x \text{ at } a &\equiv P(a) = \frac{d}{da} \text{CDF}_x(a) \\ &= \frac{d}{da} \text{CDF}_z(f^{-1}(a)) \\ &\stackrel{\boxed{\text{chain rule}}}{=} \frac{d}{du} \text{CDF}_z(u) \big|_{u=f^{-1}(a)} \frac{d}{da} f^{-1}(a) \\ &= Q(f^{-1}(a)) \frac{d}{da} f^{-1}(a)\end{aligned}$$

Change of Variable

Replacing a with x for simplicity, we have

$$P(x) = Q(f^{-1}(x)) \frac{d}{dx} f^{-1}(x)$$

As we know f is *invertible*, we can write

$$z = f^{-1}(x) \iff f(z) = x$$

Taking derivative, we have

$$\left. \begin{aligned} dz &= \frac{d}{dx} f^{-1}(x) dx \\ \dot{f}(z) dz &= dx \end{aligned} \right\} \rightsquigarrow \frac{d}{dx} f^{-1}(x) = \frac{dz}{dx} = \frac{1}{\dot{f}(z)} = \frac{1}{\dot{f}(f^{-1}(x))}$$

Change of Variable: *Scalar Result*

So, we could write

$$P(x) = \frac{Q(f^{-1}(x))}{\dot{f}(f^{-1}(x))}$$

It is easy to show that

- Having an *increasing* f is *not needed*
 ↳ For *decreasing* f , everything holds with a *sign change*
- It's though *necessary* for f to be *invertible*

Change of Variable (Scalar)

If we pass $z \sim Q(z)$ through an *invertible transform* $x = f(z)$; then,

$$x \sim P(x) = \frac{Q(f^{-1}(x))}{|\dot{f}(f^{-1}(x))|}$$

Change of Variable: Array Variables

Change of Variable (Vectorized)

Let $z \in \mathbb{R}^d$ be distributed by $z \sim Q(z)$ and $f : \mathbb{R}^d \mapsto \mathbb{R}^d$ be *invertible* and *differentiable*. Then, the transform $x = f(z)$ is distributed by

$$x \sim P(x) = \frac{Q(z)}{|\det \nabla f(z)|} \text{ with } z = f^{-1}(x)$$

1 ∇f is a $d \times d$ *Jacobian matrix*

$$\nabla f(z) = \begin{bmatrix} \partial x_1 / \partial z_1 & \cdots & \partial x_1 / \partial z_d \\ \vdots & \ddots & \vdots \\ \partial x_d / \partial z_1 & \cdots & \partial x_d / \partial z_d \end{bmatrix}$$

Change of Variable: Array Variables

Change of Variable (Vectorized)

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$$x \sim P(x) = \frac{Q(z)}{|\det \nabla f(z)|} \text{ with } z = f^{-1}(x)$$

- ② f should be *differentiable* for ∇f to exist
- ③ ∇f should be *invertible* to have

$$\det \nabla f(z) \neq 0$$

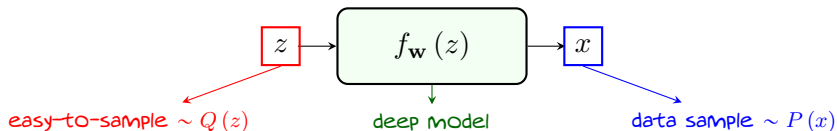
- ④ To have an *invertible* transform, z and x should be of the *same dimension*!

Flow: Learnable Mapping from Latent to Data

It is hard to learn distribution directly in the **data-space**

- It lies close to a very **thin manifold** hidden in the data-space
 ↳ Data distribution is very **complex** and the model can hardly fit to it
- Even if we learn it $\rightsquigarrow$ it's very hard **to sample from it!**

+ What if we focus on a latent space with simple distribution?



Assuming **mapping** to be deterministic and **learnable** $\rightsquigarrow$ **data distribution** is specified in terms of **this model** using change of variable

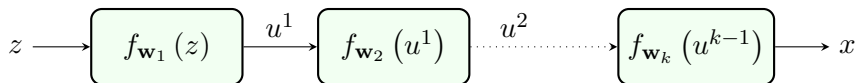
$$P_{\mathbf{w}}(x) = \frac{Q(z)}{|\det \nabla f_{\mathbf{w}}(z)|} \quad \text{with } z = f_{\mathbf{w}}^{-1}(x)$$

Flow Process: Chain of Mappings

Typically, we deal with **deep** models, i.e., we process features **sequentially**: if

- every transform in the model is **invertible** and **differentiable**

Then, distribution of each feature is given by change of variable



This describes a flow process in which

an **easy-to-sample latent representation** gradually evolves to a **data sample**

We can make this happen if we design $f_{\mathbf{w}_1}, \dots, f_{\mathbf{w}_k}$ such that

the distribution of **final output** matches the **data distribution**

Flow Process

Flow Process

Flow process describes sequential evolution of *latent* z to *data sample* x as

$$z = u^0 \xrightarrow{f_{\mathbf{w}_1}} u^1 \xrightarrow{f_{\mathbf{w}_2}} \dots \xrightarrow{f_{\mathbf{w}_{k-1}}} u^{k-1} \xrightarrow{f_{\mathbf{w}_k}} u^k = x$$

We recover latent representation of data sample x by *inverse flow* as

$$z = u^0 \xleftarrow{f_{\mathbf{w}_1}^{-1}} u^1 \xleftarrow{f_{\mathbf{w}_2}^{-1}} \dots \xleftarrow{f_{\mathbf{w}_{k-1}}^{-1}} u^{k-1} \xleftarrow{f_{\mathbf{w}_k}^{-1}} u^k = x$$

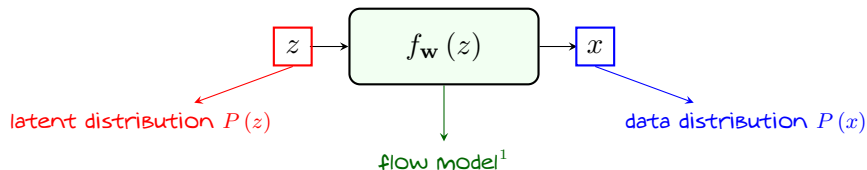
and evaluate distribution of each flow sample u^i using normalizing flow as

$$P_i(u^i) = \frac{P_{i-1}(u^{i-1})}{|\det \nabla f_{\mathbf{w}_i}(u^{i-1})|} \quad \text{with} \quad u^{i-1} = f_{\mathbf{w}_i}^{-1}(u^i)$$

Flow-based Models

Flow-based Model

A flow-based model learns a flow process which evolves a predefined latent distribution $P(z)$ to the data distribution



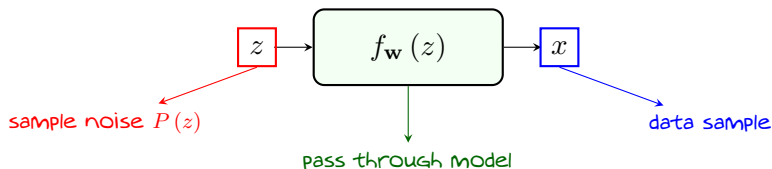
In flow-based models

- Latent distribution is an easy-to-sample one, e.g., **standard Gaussian**
- Flow model needs to be **invertible**¹

¹We represent overall deep model compactly by $f_{\mathbf{w}}$

Flow-based Models: *Sampling*

Sampling is very easy with flow-based models

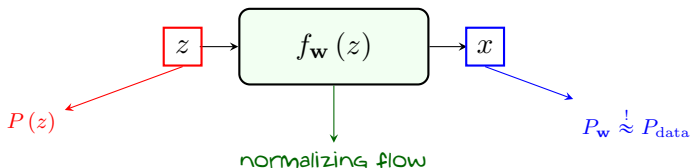


Sample_Flow():

- 1: Sample latent as $z \sim P(z)$ #Simple noise distribution
- 2: Pass through flow model $x \leftarrow f_{\mathbf{w}}(z)$
- 3: return x

Flow-based Models: MLE Learning

To train: we can maximize the likelihood of the normalized flow



For given sample $x \rightsquigarrow$ compute latent as $z_{\mathbf{w}} = f_{\mathbf{w}}^{-1}(x)$ and then likelihood as

$$P_{\mathbf{w}}(x) = \frac{P(z_{\mathbf{w}})}{|\det \nabla f_{\mathbf{w}}(z_{\mathbf{w}})|}$$

The log-likelihood is hence given by

$$\log P_{\mathbf{w}}(x) = \log P(z_{\mathbf{w}}) - \log |\det \nabla f_{\mathbf{w}}(z_{\mathbf{w}})|$$

Flow-based Models: MLE Learning

On a batch of n data samples $\{x^j \sim P_{\text{data}} : j = 1, \dots, n\}$, we have

$$\begin{aligned}
 \hat{R}(\mathbf{w}) &= -\frac{1}{n} \sum_j \log P_{\mathbf{w}}(x^j) \\
 &= \frac{1}{n} \sum_j [\log |\det \nabla f_{\mathbf{w}}(z_{\mathbf{w}}^j)| - \log P(z_{\mathbf{w}}^j)] \\
 &= \frac{1}{n} \sum_j [\log |\det \nabla f_{\mathbf{w}}(f_{\mathbf{w}}^{-1}(x^j))| - \log P(f_{\mathbf{w}}^{-1}(x^j))] \\
 &= \hat{\mathbb{E}}_{x \sim P_{\text{data}}} \{ \log |\det \nabla f_{\mathbf{w}}(f_{\mathbf{w}}^{-1}(x))| - \log P(f_{\mathbf{w}}^{-1}(x)) \}
 \end{aligned}$$

where $z_{\mathbf{w}}^j = f_{\mathbf{w}}^{-1}(x^j)$

- + Is it an easy to differentiate risk then?
- Let's take a look!

Flow-based Models: MLE Learning

Looking at the MLE risk function

$$\hat{R}(\mathbf{w}) = \hat{\mathbb{E}}_{x \sim P_{\text{data}}} \{ \log \det |\nabla f_{\mathbf{w}}(z_{\mathbf{w}})| - \log P(z_{\mathbf{w}}) \}$$

- We do know $P(z)$ $\rightsquigarrow$ we can write chain rule Backpropagation on $f_{\mathbf{w}}^{-1}$

$$\nabla_{\mathbf{w}} P(f_{\mathbf{w}}^{-1}(x)) = \nabla_z P(f_{\mathbf{w}}^{-1}(x)) \circ \boxed{\nabla_{\mathbf{w}} f_{\mathbf{w}}^{-1}(x)}$$

↳ It's easily computed by standard backpropagation

- For first term, we use $\nabla_{\mathbf{X}} \log |\det \mathbf{X}| = (\mathbf{X}^{-1})^T$ with chain rule to write

$$\nabla_{\mathbf{w}} \log |\det \nabla f_{\mathbf{w}}(f_{\mathbf{w}}^{-1}(x))| = \left(\nabla f_{\mathbf{w}}(f_{\mathbf{w}}^{-1}(x))^{-1} \right)^T \circ \nabla_{\mathbf{w}} [\nabla f_{\mathbf{w}}(f_{\mathbf{w}}^{-1}(x))]$$

↳ It needs **inverse** of Jacobian and backpropagation over Jacobian

↳ It can be computationally **expensive**, though yet **feasible**

Flow-based Models: Training

Train_Flow($\mathbb{D}$:dataset):

```

1: Initiate the flow model  $f_{\mathbf{w}}$  with some  $\mathbf{w}$ 
2: for multiple epochs do
3:   Sample a batch of data samples  $\{x^j : j = 1, \dots, n\}$  from  $\mathbb{D}$ 
4:   for  $j = 1, \dots, n$  do
5:     Invert flow to get latent representation  $z^j \leftarrow f_{\mathbf{w}}^{-1}(x^j)$ 
6:     Compute  $\hat{R}^j = \log |\det \nabla f_{\mathbf{w}}(z^j)| - \log P(z^j)$  by forward passing  $z^j$ 
7:     Backpropagate to compute  $\nabla_{\mathbf{w}} \hat{R}^j$ 
8:   end for
9:   Update  $\mathbf{w}$  using  $\text{Opt\_avg} \left\{ \nabla_{\mathbf{w}} \hat{R}^j \right\}$ 
10: end for
11: return trained flow model  $f_{\mathbf{w}}$ 

```


Real-valued Non-Volume Preserving

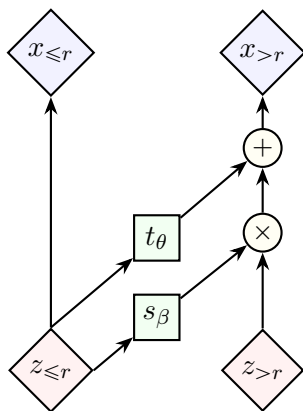
Real NVP develops *deep* flow network for *visual generation*

- It uses a *unit flow model* for *multiple steps*
 - ↳ The *inverse flow* is of a *dual* form
- It starts with *Gaussian latent*
- Flow model is designed to keep training *computationally easy*
 - ↳ It's designed to keep the derivation of

$$\hat{\mathbb{E}}_{x \sim P_{\text{data}}} \{ \log |\det \nabla f_{\mathbf{w}}(z)| \}$$

remains computationally *simple*

Real NVP: Flow



For some $r < d$, let

$$s_{\beta} : \mathbb{R}^r \mapsto \mathbb{R}^{d-r} \leftarrow \text{scaling}$$

$$t_{\theta} : \mathbb{R}^r \mapsto \mathbb{R}^{d-r} \leftarrow \text{translation}$$

be two computational models: we build flow as

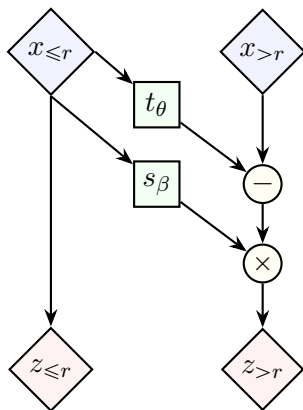
$$x_{\leq r} = z_{\leq r}$$

$$x_{> r} = z_{> r} \odot \exp \{s_{\beta}(z_{\leq r})\} + t_{\theta}(z_{\leq r})$$

whose model parameters are $\mathbf{w} = [\theta, \beta]$, i.e.,

$$x = f_{\mathbf{w}}(z)$$

Real NVP: Inverse Flow



The flow is readily inverted

$$z_{\leq r} = x_{\leq r}$$

$$z_{> r} = (x_{> r} - t_{\theta}(x_{\leq r})) \odot \exp\{-s_{\beta}(x_{\leq r})\}$$

we can compactly show it as

$$z = f_{\mathbf{w}}^{-1}(x)$$

which is of *dual* form to *direct flow* $f_{\mathbf{w}}(z)$

Real NVP: *Jacobian*

By standard derivation: we can show that

$$\nabla f_{\mathbf{w}}(\mathbf{z}) = \begin{bmatrix} \mathbf{I}_{d \times d} & \mathbf{0}_{d \times d-r} \\ \nabla_{\mathbf{x}_{\leq r}} t_{\theta}(\mathbf{x}_{\leq r}) & \begin{bmatrix} \exp\{s_1\} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \exp\{s_{d-r}\} \end{bmatrix} \end{bmatrix}$$

where $s_1, \dots, s_{d-r}$ are the outputs of scaling model, i.e.,

$$\{s_1, \dots, s_{d-r}\} \leftarrow s_{\beta}(\mathbf{x}_{\leq r})$$

This is an upper-triangle matrix whose determinant $\equiv$ product of main diagonal

$$|\det \nabla f_{\mathbf{w}}(\mathbf{z})| = \prod_{j=1}^{d-r} \exp\{s_j\} = \exp\left\{\sum_j s_j\right\}$$

Real NVP: Training

As a result

$$\log |\det \nabla f_{\mathbf{w}}(z)| = \sum_j s_j = \text{sum}[s_{\beta}(x_{\leq r})]$$

whose gradient is readily given by

$$\nabla_{\mathbf{w}} \log |\det \nabla f_{\mathbf{w}}(z_{\mathbf{w}})| = \text{sum}[\nabla_{\beta} s_{\beta}(x_{\leq r})]$$

Also, if we consider *standard Gaussian latent*, we have

$$\log P(z) = -\frac{1}{2}\|z\|^2 - \text{Constant}$$

and thus, the gradient is readily computed by backpropagation

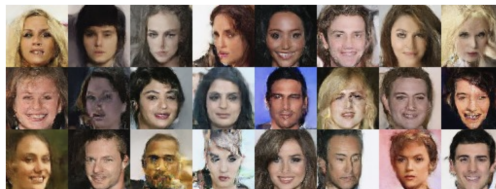
$$\nabla_{\mathbf{w}} \log P(z) = -\frac{1}{2} \nabla \|f_{\mathbf{w}}^{-1}(x)\|^2$$

Real NVP: *Sample Output*

Data samples from CelebA



Samples from the flow network



Variants of Real NVP: NICE

Real NVP was indeed the extension of *an initial architecture*

NICE: Non-linear Independent Component Estimation (NICE)

which mainly had a simpler flow

$$\begin{aligned}x_{\leq r} &= z_{\leq r} \\x_{> r} &= z_{> r} + t_{\theta}(z_{\leq r})\end{aligned}$$

This model was one of earliest computational flow-based models

✓ It was simple to train due to simple risk function

✗ It was not too expressive, i.e., **limited in capacity**

The **latter issue** was the reason for Real NVP proposal

Variants of Real NVP: Glow

Real NVP was later extended to a more flexible architecture called

Glow: Generative Flow

in which *permutation* of data entries, i.e., sorting of x_i 's in x , is *learned* as well

- Glow uses *convolution to combine entries* in the flow
 - ↳ This enables the possibility to *learn how to couple channels*
 - ↳ Remember that in Real NVP, the coupling is *fixed*
 - ↳ We fix choice of r and combine $x_{\leq r}$ with $x_{>r}$
- The training may need *more computation*
- Generation quality is generally *better*

Some other known extensions are

- FFJORD, Flow++, VFlow, Wavelet Flow, Flow Matching, Rectified Flow

Flow-based Models: Wrap Up

Flow-based models sound classic: they are

- *easy to sample* $\rightsquigarrow$ pass *noise* sample $z \sim P(z)$ through flow network
 - ↳ Faster than AR models
 - ↳ Both faster and easier than EBMs
- imposing *high computation* for training, as we need to solve

$$\min_{\mathbf{w}} \mathbb{E}_{x \sim P_{\text{data}}} \{ \log |\det \nabla f_{\mathbf{w}}(z)| - \log P(z) \} \quad \text{with } z = f_{\mathbf{w}}^{-1}(x)$$

which can be computationally *expensive*

- ↳ Generally, more complex than classic AR models
- ↳ Definitely *easier than EBMs*

Final Notes

We are over with **explicit** methods: *in all these methods*

we **directly** targeted to learn **data distribution**

These three methods, i.e., AR, EBM and Flow-based make the foundation

Make More Explicit Approaches!

We can **combine** these methods to come up with other approaches

- In **AR flow-based** modeling, we learn **autoregressive** conditional distributions by **flow-based models**

Though effective, these approaches limit us in various senses

! In flow-based models we had to use **same-dimension latent space**

↳ We know that we can have **much smaller latent** space!

We **next** study approaches in which we learn data distribution **implicitly!**

Homework: Investigate Further!

Assume that we have *latent* $z \in \mathbb{R}^m$ with

$$z \sim Q(z)$$

We pass it through *function* $f : \mathbb{R}^m \mapsto \mathbb{R}^d$ for $d > m$ and compute x

$$x = f(z)$$

Obviously, f is *not invertible*

Question

How can we write $P(x)$ in terms of $Q(z)$ and f ?